

Table 2

|          |                   | ReLU                      |                           |                |                       |  | Tanh                      |                           |                |                       |
|----------|-------------------|---------------------------|---------------------------|----------------|-----------------------|--|---------------------------|---------------------------|----------------|-----------------------|
| data set | training set size | Original hyper-parameters | Modified hyper-parameters | accuracy-paper | accuracy-reproduction |  | Original hyper-parameters | Modified hyper-parameters | accuracy-paper | accuracy-reproduction |
| mnist    | 100               | GP-100-1.79-0.83          |                           | 0.7735         | 0.7735                |  | GP-100-3.14-0.97          |                           | 0.7736         | 0.7736                |
|          | 200               | GP-100-1.79-0.83          |                           | 0.8282         | 0.8282                |  | GP-100-3.99-2.00          |                           | 0.8277         | 0.8277                |
|          | 500               | GP-100-1.79-0.83          |                           | 0.8995         | killed                |  | GP-50-3.48-1.86           |                           | 0.9008         | 0.9008                |
|          |                   |                           | GP-68-1.79-0.83           |                | 0.8995                |  |                           |                           |                |                       |
|          |                   |                           | GP-50-1.75962-0.859       |                | 0.8995                |  |                           |                           |                |                       |
|          |                   |                           | GP-20-1.745-0.7           |                | 0.8995                |  |                           |                           |                |                       |
|          | 1000              | GP-20-1.45-0.28           |                           | 0.9279         | 0.9279                |  | GP-20-1.96-0.62           |                           | 0.9266         | 0.9266                |
|          | 2000              | GP-10-1.11-0.55           |                           | 0.9485         | 0.9485                |  | GP-10-1.79-1.45           |                           | 0.9477         | 0.9477                |
|          | 5000              | GP-7-0.61-0.07            |                           | 0.9692         | 0.9692                |  | GP-3-1.11-0.00            |                           | 0.9693         | 0.9693                |
|          | 10000             | GP-7-0.61-0.07            |                           |                | killed                |  | GP-2-1.62-0.28            |                           | 0.9773         | killed                |
|          | 20000             | GP-5-1.62-0.83            |                           |                | killed                |  | GP-1-2.63-0.00            |                           | 0.9836         | killed                |
|          | 50000             | GP-1-0.10-0.48            |                           |                | killed                |  | GP-1-1.28-0.00            |                           | 0.9879         | killed                |
| cifar10  | 100               | GP-3-4.49-0.97            | GP-3-5-0.05               | 0.2673         | 0.2203                |  | GP-10-3.65-1.17           | GP-6-3.56-4.5             | 0.2718         | 0.2171                |
|          | 200               | GP-3-3.99-1.72            | GP-3-4.99-1.9             | 0.3022         | 0.2793                |  | GP-7-3.65-0.55            | GP-8-4.16-4.97            | 0.2927         | 0.2751                |
|          | 500               | GP-20-1.79-0.21           | GP-20-1.53-0.489          | 0.3395         | 0.3395                |  | GP-7-3.65-0.62            | GP-5-3.4-0.22             | 0.3291         | 0.3291                |
|          | 1000              | GP-7-1.28-0.00            | GP-7-1.28-0.54            | 0.3608         | 0.3608                |  | GP-50-2.97-0.97           | GP-5-2.97-0.74            | 0.3702         | 0.3702                |
|          | 2000              | GP-3-4.16-1.17            | GP-3-2.12-0.25            | 0.3953         | 0.3953                |  | GP-5-4.66-1.03            | GP-5-4.56-0.054           | 0.3959         | 0.3958                |
|          | 5000              | GP-3-4.66-1.03            | GP-3-2.15-2.025           | 0.4454         | 0.4454                |  | GP-10-3.65-1.38           | GP-3-3.67-0.05            | 0.443          | 0.443                 |
|          | 10000             | GP-5-2.97-0.28            |                           | 0.478          | killed                |  | GP-7-3.48-2.00            |                           | 0.4766         | killed                |
|          | 20000             | GP-3-5.00-0.83            |                           | 0.5011         | killed                |  | GP-7-3.14-1.93            |                           | 0.5124         | killed                |
|          | 45000             | GP-3-3.31-1.86            |                           | 0.5566         | killed                |  | GP-3-3.48-1.52            |                           | 0.5558         | killed                |